

Quantile Regression

Mathematical Formulation and Real World Uses

Giacomo Boschi

September 9th, 2026

Table of Contents

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

1 The Concept

2 Quantile Regression

3 MATLAB Solution

4 Results

5 Final Remarks

The Foundation

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

In OLS, the parameters (β_0, β_1) are obtained by minimizing the sum of squared residuals, which estimates the conditional mean $E(Y|X)$:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

If we replace the squared loss with the absolute value, we get the LAD, which estimates the conditional median $Q_{0,5}(Y|X)$:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n |y_i - (\beta_0 + \beta_1 x_i)|$$

From Median to Quantiles through optimization

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

The conditional median is a specific quantile; can the loss function be modified to target other quantiles?

The absolute value is symmetric: it penalises positive and negative residuals equally, this is why we get the median. Therefore to target other quantiles, we have to give different weights to positive and negative residuals:

$$\rho_{\tau}(\epsilon) = \begin{cases} \tau\epsilon & \text{if } \epsilon \geq 0 \\ (\tau - 1)\epsilon & \text{if } \epsilon < 0 \end{cases} \quad \text{with } \tau \in (0, 1)$$

where τ is the targeted quantile.

The Optimization Problem

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Replacing the absolute value with ρ_τ in the regression problem, we obtain the quantile regression, a generalisation of the LAD, which estimates the conditional quantile $Q_\tau(Y|X)$

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n \rho_\tau(y_i - (\beta_0 + \beta_1 x_i))$$

Just like the LAD, this function is piecewise linear and can't be differentiable at 0; one of the approaches is to reformulate and solve it as an LP problem

The LP Reformulation: splitting the residuals

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Each residual $\epsilon = y_i - (\beta_0 + \beta_1 x_i)$ can be decomposed into its positive and negative parts:

$$\epsilon_i = u_i - v_i, \quad \text{with } u_i, v_i \geq 0$$

$$\left\{ \begin{array}{ll} \epsilon_i \geq 0: & u_i = \epsilon_i, v_i = 0 \\ & \Rightarrow \rho_\tau(\epsilon_i) = \tau \epsilon_i = \tau u_i + (1 - \tau) \cdot 0 \\ \epsilon_i < 0: & u_i = 0, v_i = -\epsilon_i \\ & \Rightarrow \rho_\tau(\epsilon_i) = (\tau - 1) \epsilon_i = \tau \cdot 0 + (1 - \tau) v_i \end{array} \right.$$

In both cases: $\rho_\tau(\epsilon_i) = \tau u_i + (1 - \tau) v_i$

The LP Reformulation: complete problem

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Therefore the complete reformulation of the problem can be summarized in this form:

$$\begin{aligned} \min_{\beta_0, \beta_1, u_i, v_i} \quad & \tau \sum_{i=1}^n u_i + (1 - \tau) \sum_{i=1}^n v_i \\ \text{s.t.} \quad & y_i - (\beta_0 + \beta_1 x_i) = u_i - v_i \quad \forall i = 1, \dots, n \\ & u_i, v_i \geq 0 \quad \forall i = 1, \dots, n \end{aligned}$$

In this form the problem is an LP, which is computationally trivial, and directly translatable to the solver

YALMIP and Gurobi Implementation

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

```
beta = sdpvar(p, 1);
u = sdpvar(n, 1);
v = sdpvar(n, 1);
objective = tau * sum(u) + (1-tau) * sum(v);
|
options = sdpsettings('solver', 'gurobi', 'verbose', 0);
|
C = [y == X * beta + u - v;
     u >= 0;
     v >= 0];
|
sol = optimize(C, objective, options);
|
if sol.problem ~= 0
|   error("Error! The solver didn't find an optimal solution")
end
|
beta_hat = value(beta);
|
```


Engel Curves via Quantile Regression

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

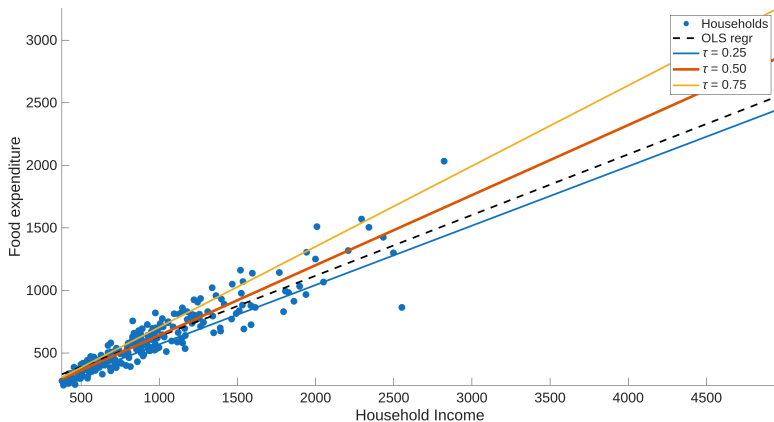


Figure: Engel Curves Dataset (Koenker & Hallock 2001, Fig. 3)

OLS vs Quantile Regression

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

OLS

- Estimates the conditional mean
- Sensitive to outliers (squared loss)
- One optimization problem, one slope
- Can't capture how the relationship varies across the distribution

Quantile Regression

- Estimates any conditional quantile
- Less sensitive to outliers, like LAD
- Family of LPs in τ , multiple slopes
- Naturally reveals how the relationship changes across quantiles

Advantages, Limitations and Applications

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Advantages

- Less sensitive to outliers than other methods
- Naturally reveals the relationship across different quantiles
- Requires no distributional assumption on the errors

Limitations

- More computationally expensive than linear regression
- The extra robustness is sometimes unnecessary

Applications

Suited to any setting where the conditional distribution of $Y|X$ varies in shape across X

- **Demand Analysis:** e.g. Engel Curves (Fig. 1) or Alcohol Demand
- **Labor economics:** e.g. Wage Inequality
- **Medical Research:** e.g. Birthweight (Koenker & Hallock, 2001)

“Just as the mean gives an incomplete picture of a single distribution, so the regression curve gives a corresponding incomplete picture for a set of distributions.”

— Mosteller & Tukey (1977)

“[Quantile regression is] a comprehensive strategy for completing the regression picture.”

— Koenker & Hallock (2001)

References

Quantile
Regression

Giacomo
Boschi

The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

- [1] Koenker, R. and Bassett, G. (1978). *Regression Quantiles*. *Econometrica*, 46(1), 33–50.
- [2] Koenker, R. and Hallock, K.F. (2001). *Quantile Regression*. *Journal of Economic Perspectives*, 15(4), 143–156.
- [3] Mosteller, F. and Tukey, J. (1977). *Data Analysis and Regression: A Second Course in Statistics*. Addison-Wesley. As cited in Koenker & Hallock (2001).
- [4] Lecture presentations on OLS and LAD regression, Optimization Fundamentals Course.